

## Newton-Raphson

**input** : function  $f$  and its derivative  $f'$ , initial guess  $x_0$ ,  
tolerance  $\varepsilon_s$ , maximum iterations  $N$

**output:** estimate  $x_r$  of the root

$x_r \leftarrow x_0$

**for**  $i \leftarrow 1$  **to**  $N$  **do**

$x_{r,\text{old}} \leftarrow x_r$

**if**  $f'(x_{r,\text{old}}) = 0$  **then**

**return** failure // horizontal tangent

**end**

    // zero of the tangent line at  $x_{r,\text{old}}$

$x_r \leftarrow x_{r,\text{old}} - \frac{f(x_{r,\text{old}})}{f'(x_{r,\text{old}})}$

$\varepsilon_a \leftarrow \left| \frac{x_r - x_{r,\text{old}}}{x_r} \right| \cdot 100\%$

**if**  $\varepsilon_a < \varepsilon_s$  **then**

**return**  $x_r$

**end**

**end**

**return**  $x_r$  // did not converge in  $N$  iterations